LUMAFLOW · DESIGN DECISION RECORD

LumaFlow TRL 2 review decisions

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DOCUMENT CONTROL

Document ID	LMF-DDR-001	Version	0.3
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Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/lumaflow		
Source	docs/decisions/0001-trl2-review-decisions.md @ f88a9c3		

REVISION HISTORY

Version	Date	Author	Change
0.3	2026-10-02	Amish Chadha	O1 decided by Amish as recommended (LMF-DEC-001)
0.2	2026-09-25	Amish Chadha	Recommendations accepted by Amish (DDR-002); O2 and O3 decided, O1 stays open
0.1	2026-09-25	Amish Chadha	Record Amish's decisions on the TRL 2 review items and the items that remain open

0001: TRL 2 review decisions

- **Date:** 2026-09-25
- **Status:** accepted for items D1 to D9; O2 and O3 accepted in LMF-DDR-002; O1 decided by Amish on 2026-10-02: "i approve your recommendations for all 555 open decisions." (LMF-DEC-001)

Context

The TRL 2 review note (`docs/REVIEW.md`, session 2026-09-25, /populate) listed nine items as "Proposed, awaiting Amish". On 2026-09-25 Amish reviewed the review points for every portfolio repo and wrote: "proceed with all of your recommendations across all batches. Make sure we don't proceed to TRL 4 on any of them." Every item that carried a recommendation is therefore decided in favor of that recommendation. Items without a recommendation stay open. TRL 4 work is on hold by the same instruction.

Options considered

The options for each item are those listed in `docs/REVIEW.md` (TRL 2 session) and in LMF-PRC-001 v0.2, Key design choices.

Decision

Table 1. Decided items.

#	Item	Decision
D1	Design water quality and dose target	Option A: design for 90 %/cm UVT, alarm and close the valve below 40 mJ/cm², and keep R3 (70 %/cm) visibly not met; Option C (flow limiting in poor water) is a later upgrade. Decided by Amish, 2026-09-25: go with recommendation.
D2	Budget	Raise `budget_usd` from $200 to $225, keeping the shutoff valve as the fail-safe. Decided by Amish, 2026-09-25: go with recommendation.
D3	LED layout	Axial LED head shining up the tube through a flat quartz window, rather than LEDs around a quartz sleeve. Decided by Amish, 2026-09-25: go with recommendation.
D4	Wavelength	275 nm LEDs rather than 265 nm, prices to be revisited. Decided by Amish, 2026-09-25: go with recommendation.
D5	Flow switch	Hall-effect flow sensor as both switch and meter, rather than a reed flow switch. Decided by Amish, 2026-09-25: go with recommendation.
D6	Fail-safe	Normally closed outlet valve that closes on alarm and on power loss. Decided by Amish, 2026-09-25: go with recommendation.
D7	Orientation	Vertical mounting with upward flow and the LEDs at the bottom. Decided by Amish, 2026-09-25: go with recommendation.
D8	Power	External certified 24 V adapter on a GFCI or RCD-protected outlet; only 24 V DC at the unit. Decided by Amish, 2026-09-25: go with recommendation.
D9	First user	A well-water household with a sediment pre-filter. Decided by Amish, 2026-09-25: go with recommendation.

Notes on the decided items:

- **D1.** R2 and R3 in LMF-REQ-001 v0.3 keep their targets. LMF-CAL-001 shows that the TRL 2 dose estimate was about three times too high: at 90 %/cm the laminar RED is 14.7 mJ/cm² [B5], so R2 is now not met as well as R3. Under D1 the unit, as drawn, would alarm and close its valve in any water. The routes to close R2 are new proposals (O3) in `docs/REVIEW.md`; D1 itself is recorded as decided and not reopened here.
- **D2.** `budget_usd` in `project.yaml` is now 225, and R16 is redefined to $225. The TRL 3 BOM totals $249.00 [I1], so R16 is still not met; see O3.
- **D5.** At TRL 3 the assumed sensor gives only 0.04 s of margin on the 0.5 s switch-on time (LMF-CAL-001, D2); a higher pulse-rate sensor is proposed in O3. The decision to use a Hall-effect sensor stands.
- **SwapCell.** LumaFlow is mains-powered through a 24 V adapter and uses no SwapCell pack. The portfolio decisions of 2026-09-25 on the SwapCell interface (v0.3 items: wake without CAN, charge-while-discharging mode, latch vibration rating) and on pricing shared packs once do not change the LumaFlow design or BOM.
- **Pitch and problem lines.** The review recommended no change to the `pitch` or `problem` wording, so `project.yaml` and `README.md` keep them.

Table 2. Items left open by this record. O2 and O3 were decided on 2026-09-25 in LMF-DDR-002; O1 stays open.

#	Item	Why it stays open
O1	Partner that can supply real well-water UVT data and later test the reactor (a well-water association, a university lab or a water charity)	No partner was recommended at TRL 2. Decided by Amish, 2026-10-02, as later recommended: a university extension programme for private well owners, with the Texas A&M AgriLife Extension Texas Well Owner Network as the first candidate to approach (LMF-DEC-001)
O2	End cap material (acetal or 316 stainless)	Listed as an open question at TRL 2 with no recommendation. The TRL 3 model uses a 316 lower cap and a PTFE-shielded acetal upper cap, which LMF-CAL-001 supports (F5 and R11). Decided by Amish, 2026-09-25: go with recommendation (confirm), LMF-DDR-002 E1
O3	New TRL 3 proposals: route to close R2, dose monitor position, budget after re-pricing, flow sensor pulse rate, pressure limiter as an installation requirement, thermal cut-back	Raised by LMF-CAL-001 after the decision. Decided by Amish, 2026-09-25: go with recommendation, LMF-DDR-002 E2 to E8 (the budget figure after re-pricing stays open)

Consequences

- LMF-PRB-001, LMF-PRC-001 and LMF-REQ-001 are revised to v0.3 to show the decisions: the design choices D1 and D3 to D8 are no longer "proposed", the first user is named, R16 carries the $225 target, and the numbers are replaced by those of LMF-CAL-001.
- `project.yaml` carries `budget_usd: 225`.
- The partner question (O1) stayed open in LMF-PRB-001 until Amish decided it on 2026-10-02 (LMF-DEC-001).